A PROJECT REPORT

on

Ayurvedic Classification as a Rule-Based Expert System

Submitted By

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CERTIFICATE

This is to certify that PANDEY SHUBHAMKUMAR of T.Y.B.Sc. Computer Science **(Sem V)** class has satisfactorily completed the Project **"Ayurvedic Classification as a Rule-Based Expert System"** to be submitted in the partial fulfillment for the award of **Bachelor of Science** in **Computer Science** during the academic year 2026-2027.

Date of Submission:

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DECLARATION

We, **PANDEY SHUBHAMKUMAR** hereby declare that the project entitled **"Ayurvedic Classification as a Rule-Based Expert System "** Submitted in the partial fulfillment for the award of **Bachelor of Science** in **Computer Science** during the academic year **2026-27** is my original work and the project has not formed the basis for the award of any degree, associateship, fellowship or any other similar titles.

Signature of the Student:

Place:

Date:

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CHAPTER 1: INTRODUCTION

1.1 Background

India has a rich tradition of knowledge covering medicine, mathematics, astronomy, philosophy, architecture and many other fields. This body of knowledge is called the Indian Knowledge System (IKS). One of its most important branches is Ayurveda, a holistic system of health care that is more than 3,000 years old. Ayurveda does not treat every person in the same way; it first understands the natural constitution of the individual, called Prakriti, and then suggests diet and lifestyle suited to that constitution.

Computer Science, especially Artificial Intelligence, offers techniques for capturing the knowledge of a human expert in a computer program. An expert system is one such technique. Combining the two, this project builds a small expert system that behaves like a simple Vaidya assistant for classifying Prakriti.

1.2 Problem Statement

The classification of Prakriti normally depends on the experience of a Vaidya. The classical rules are scattered across texts and are difficult for a beginner to apply consistently. There is a need for a simple, transparent and repeatable system that applies the same rules to every person and explains how it arrived at its result.

1.3 Objectives

- To study the concepts of Ayurveda related to Tridosha and Prakriti as part of the Indian Knowledge System.
- To represent the classification knowledge as IF-THEN rules in a knowledge base.
- To design an inference engine that applies the rules using forward chaining.
- To classify a person into one of the seven Prakriti types.
- To provide an explanation of the result by displaying the rules that were fired.
- To implement and test the system using Python.

1.4 Scope

The system classifies Prakriti only. It uses eight characteristics which the user can answer without any medical equipment. It gives general diet and lifestyle guidance taken from the classical principles. It does not diagnose diseases and does not prescribe medicines.

1.5 Relevance to Indian Knowledge System

The project preserves and presents a traditional Indian classification method in a digital form. It shows that classical knowledge is logical and structured, and that it can be studied, tested and extended using modern computing tools.

1.6 Tools and Technologies

Table 1.1: Tools and technologies used

Item	**Details**
Programming language	Python 3
Technique	Rule-based expert system with forward chaining
Interface	Command line (text based question and answer)
Editor / IDE	IDLE, VS Code or any Python editor
Operating system	Windows / Linux / macOS

CHAPTER 2: AYURVEDA AND THE INDIAN KNOWLEDGE SYSTEM

2.1 Indian Knowledge System

The Indian Knowledge System refers to the knowledge developed in India over thousands of years, passed on through texts and teacher-student traditions. It includes the Vedas, Upanishads, Yoga, Ayurveda, Sulba Sutras, Aryabhatiya and many other works. Its main idea is that knowledge should be connected to practical life and to the well-being of people and nature.

2.2 Ayurveda

The word Ayurveda comes from two Sanskrit words: *Ayu* (life) and *Veda* (knowledge), so it means "the science of life". It is regarded as an Upaveda of the Atharvaveda. The three major classical texts, known as the Brihat Trayi, are:

- **Charaka Samhita** - mainly deals with internal medicine and principles of health.
- **Sushruta Samhita** - mainly deals with surgery and anatomy.
- **Ashtanga Hridaya** (Vagbhata) - a summary of the above two works.

Ayurveda defines health as a state of balance between the doshas, proper digestion and metabolism, healthy tissues and waste elimination, together with a pleasant state of mind, senses and soul.

2.3 Panchamahabhuta (Five Elements)

According to Ayurveda everything in the universe, including the human body, is made of five basic elements: **Akasha** (space), **Vayu** (air), **Agni** (fire), **Jala** (water) and **Prithvi** (earth). Different combinations of these elements form the three doshas.

2.4 Tridosha

The three doshas are the functional principles that control the working of the body and mind. **Vata** governs movement, **Pitta** governs transformation and digestion, and **Kapha** governs structure and lubrication. Good health exists when the doshas are in balance; imbalance leads to disorders.

2.5 Prakriti

Prakriti is the natural constitution of an individual, determined at the time of conception and birth by the dominance of one or more doshas. It remains the same throughout life and decides a person's physical build, digestion, mental tendencies and the way the body responds to food, weather and stress. Knowing the Prakriti helps in preventive care and in planning personalised diet and lifestyle.

CHAPTER 3: AYURVEDIC CLASSIFICATION

3.1 Concept of Classification in Ayurveda

Ayurveda classifies doshas, tissues (dhatus), tastes (rasa), foods, medicinal plants and body types. Each class has known properties, so once an item is placed in a class, its behaviour can be predicted. This project focuses on classifying individuals into Prakriti types.

3.2 Characteristics of the Three Doshas

Table 3.1: Characteristics of Vata, Pitta and Kapha

Feature	Vata	Pitta	Kapha
Elements	Air + Space	Fire + Water	Water + Earth
Main qualities	Dry, light, cold, mobile	Hot, sharp, oily, liquid	Heavy, slow, cool, steady
Main seat	Colon	Small intestine, stomach	Chest, stomach
Main functions	Movement, breathing, nerve impulses, elimination	Digestion, metabolism, body heat, vision	Structure, stability, lubrication, immunity
Tendency when imbalanced	Anxiety, dryness, constipation, poor sleep	Anger, acidity, inflammation, skin rashes	Lethargy, weight gain, congestion, oversleeping

3.3 Types of Prakriti

Classical texts describe seven types of Prakriti depending on which dosha or doshas dominate at birth:

Table 3.2: Seven types of Prakriti

Group	Prakriti type	Description
Ekadoshaja (single dosha)	Vata, Pitta, Kapha	One dosha is clearly dominant
Dvandvaja (dual dosha)	Vata-Pitta, Pitta-Kapha, Vata-Kapha	Two doshas are almost equally dominant
Sannipataja / Sama (all three)	Vata-Pitta-Kapha	All three doshas are in nearly equal proportion

3.4 Why Classification Helps

- Personalised diet and daily and seasonal routines.
- Knowing which health problems a person is prone to.
- Preventive care before an imbalance becomes a disease.

CHAPTER 4: EXPERT SYSTEMS

4.1 Introduction

An expert system is a computer program that stores the knowledge of a human expert in a particular field and uses it to solve problems or give advice in that field. It is a major application area of Artificial Intelligence. Well-known early examples are MYCIN (medical diagnosis) and DENDRAL (chemical analysis).

4.2 Components of an Expert System

- **Knowledge Base** - stores the facts and rules of the domain.
- **Working Memory** - stores the facts about the current case.
- **Inference Engine** - applies the rules to the facts to derive new conclusions.
- **User Interface** - collects input from the user and shows the result.
- **Explanation Facility** - explains how a conclusion was reached.

4.3 Rule-Based Systems

In a rule-based expert system, knowledge is written as production rules of the form **IF** condition **THEN** action/conclusion. Example: *IF skin = dry THEN Vata score + 1*. Rules are simple to write, easy to read and easy to modify without changing the program.

4.4 Forward Chaining and Backward Chaining

Table 4.1: Forward vs backward chaining

Forward chaining	**Backward chaining**
Starts from known facts and moves towards a conclusion (data driven).	Starts from a goal and works backwards to find supporting facts (goal driven).
Suitable when all inputs are collected first.	Suitable when the goal is known and facts are searched on demand.
Used in this project.	Not used in this project.

4.5 Why a Rule-Based System Suits Ayurveda

- The classical texts already describe knowledge as observation-to-dosha relationships, which map directly to IF-THEN rules.
- The reasoning is transparent: every result can be traced to the rules that produced it.
- New rules can be added by a domain expert without rewriting the program, and no large training data is needed, unlike machine learning.

CHAPTER 5: SYSTEM ANALYSIS AND DESIGN

5.1 Problem Analysis

Input: answers to eight questions about the person's body and behaviour.
Process: match answers with the rules, score each dosha, then apply the classification rules.
Output: dosha scores, Prakriti type, general diet/lifestyle guidance and the list of rules fired.

5.2 System Architecture

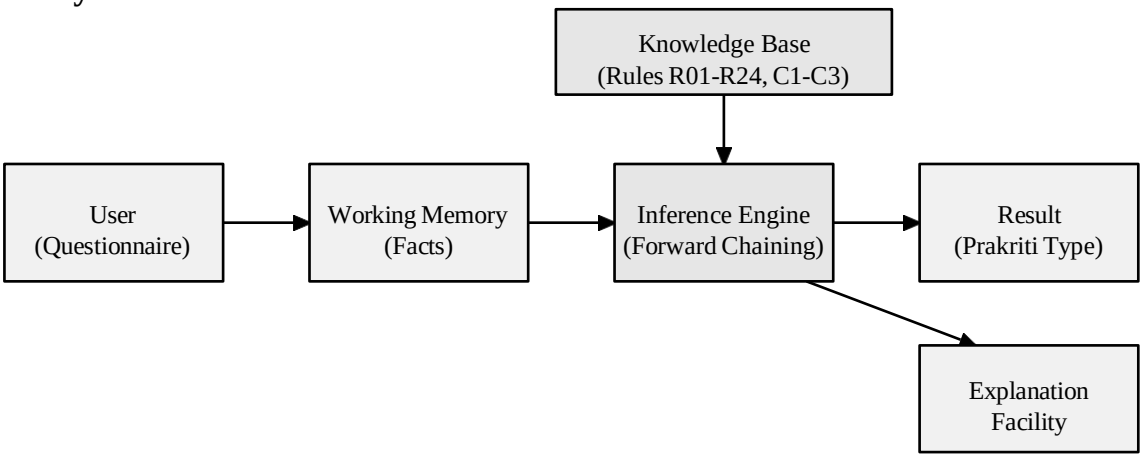


Figure 5.1: Architecture of the Ayurvedic rule-based expert system

5.3 Knowledge Base: Attributes

The knowledge base uses eight attributes. For each attribute there are three possible values, one pointing to each dosha.

Table 5.1: Attributes and their values

Attribute	Vata value	Pitta value	Kapha value
Body frame	Thin and light	Medium and muscular	Large and heavy
Skin type	Dry and rough	Warm, sensitive, prone to redness	Oily, smooth and soft
Appetite and digestion	Irregular appetite	Strong appetite, quick digestion	Low appetite, slow digestion
Sleep pattern	Light, easily disturbed	Moderate and sound	Deep and long
Climate dislike	Dislikes cold and wind	Dislikes heat and sun	Dislikes damp, cool weather

Temperament	Quick, enthusiastic, anxious	Sharp, focused, irritable	Calm, steady, forgiving
Activity level	Restless, always moving	Purposeful and goal oriented	Slow, steady, enduring
Reaction to stress	Worry and fear	Anger and frustration	Withdrawal and silence

5.4 Knowledge Base: Rules

Scoring rules (R01 to R24). Each rule has the form IF attribute = value THEN add 1 to the matching dosha. Since there are 8 attributes, every person gets a total score of 8 distributed among the three doshas.

Table 5.2: Scoring rules

Rule	IF (condition)	THEN (action)
R01	frame = thin	Vata score + 1
R02	frame = medium	Pitta score + 1
R03	frame = large	Kapha score + 1
R04	skin = dry	Vata score + 1
R05	skin = warm	Pitta score + 1
R06	skin = oily	Kapha score + 1
R07	appetite = irregular	Vata score + 1
R08	appetite = strong	Pitta score + 1
R09	appetite = slow	Kapha score + 1
R10	sleep = light	Vata score + 1
R11	sleep = moderate	Pitta score + 1
R12	sleep = deep	Kapha score + 1
R13	climate = dislikes_cold	Vata score + 1
R14	climate = dislikes_heat	Pitta score + 1
R15	climate = dislikes_damp	Kapha score + 1

R16	temperament = quick_anxious	Vata score + 1
R17	temperament = sharp_irritable	Pitta score + 1
R18	temperament = calm_steady	Kapha score + 1
R19	activity = restless	Vata score + 1
R20	activity = purposeful	Pitta score + 1
R21	activity = slow_steady	Kapha score + 1
R22	stress = worry	Vata score + 1
R23	stress = anger	Pitta score + 1
R24	stress = withdrawal	Kapha score + 1

Classification rules. After scoring, let *top*, *mid* and *low* be the highest, middle and lowest dosha scores.

Table 5.3: Classification rules

Rule	IF (condition)	THEN (conclusion)
C1	top - mid >= 2	Single dosha Prakriti of the top dosha (Ekadoshaja)
C2	top - mid < 2 AND top - low >= 2	Dual dosha Prakriti of the top two doshas (Dvandvaja)
C3	top - low <= 1	Sama Prakriti, all three doshas balanced (Sannipataja)

Rules are checked in the order C1, C2, C3 and the first rule whose condition is true is fired.

5.5 Inference Engine: Algorithm

- **Step 1:** Start and display the questions one by one.
- **Step 2:** Store each answer as a fact in the working memory.
- **Step 3:** For every scoring rule R01-R24, check whether its condition matches the facts; if yes, fire it, add 1 to the dosha and record the rule in the trace.
- **Step 4:** Sort the three doshas by score and find top, mid and low.

- **Step 5:** Check C1, C2 and C3 in order and fire the first one that is true.
- **Step 6:** Display the scores, Prakriti type and general guidance.
- **Step 7:** Display the trace of fired rules as the explanation. Stop.

5.6 Explanation Facility

Every fired rule is recorded with its rule number and reason. After the result, the user can see exactly which answers contributed to which dosha. This makes the system transparent and also useful for learning, as a student can verify the result manually.

CHAPTER 6: IMPLEMENTATION

6.1 Implementation Environment

The system is implemented as a single Python file, `expert_system.py`, which needs only the standard Python 3 installation. No external library is required. The complete source code is given in Appendix A.

6.2 Modules of the Program

Table 6.1: Modules of the program

Module / function	Purpose
ATTRIBUTES	Dictionary holding the eight questions, their three possible values and descriptions (knowledge base: facts).
build_rules()	Generates the 24 scoring rules R01-R24 from ATTRIBUTES (knowledge base: rules).
infer(facts)	Inference engine. Fires matching rules, calculates scores, applies classification rules C1-C3 and returns result, advice and trace.
ask_user()	User interface. Asks the questions and stores the answers as facts.
show(out)	Displays scores, Prakriti type, advice and the explanation.

6.3 Core Logic

The heart of the system is the function `infer()`. It first loops through all scoring rules and fires those that match the facts. It then ranks the doshas and applies the classification rules:

```
    for    r    in    RULES:     if
facts.get(r["attr"]) == r["val"]:
            score[r["dosha"]] += 1
trace.append(...)      #   record   fired   rule   ranked   =
sorted(DOSHAS, key=lambda d: -score[d])    top, mid, low =
(score[d] for d in ranked)     if top - mid >= 2:
# Rule C1          result = ranked[0] + " Prakriti
(Ekadoshaja)"     elif top - low <= 1:               # Rule
C3          result = "Sama Prakriti (Sannipataja)"     else:
# Rule C2
        result = ranked[0] + "-" + ranked[1] + " Prakriti (Dvandvaja)"
```

6.4 How to Run

- Install Python 3 on the computer.
- Save the program as `expert_system.py`.
- Open a terminal in that folder and run `python expert_system.py`.
- Answer the eight questions by entering 1, 2 or 3.

CHAPTER 7: TESTING AND RESULTS

7.1 Test Cases

Eleven test cases cover single-dosha, dual-dosha and balanced outcomes, decision boundaries and invalid input. The scores were produced by running the program.

Table 7.1: Test cases and results (input order: frame, skin, appetite, sleep, climate, temperament, activity, stress)

ID	Test input (frame, skin, appetite, sleep, climate, temperament, activity, stress)	Scores (V/P/K)	Expected result	Rule fired	Status
T1	thin, dry, irregular, light, dislikes_cold, quick_anxious, restless, anger	7 / 1 / 0	Vata Prakriti (Ekadoshaja)	C1	Pass
T2	medium, warm, strong, moderate, dislikes_heat, sharp_irritable, purposeful, worry	1 / 7 / 0	Pitta Prakriti (Ekadoshaja)	C1	Pass
T3	large, oily, strong, moderate, dislikes_damp, sharp_irritable, slow_steady, anger	0 / 4 / 4	Pitta-Kapha Prakriti (Dvandvaja)	C2	Pass
T4	thin, warm, slow, light, dislikes_heat, calm_steady, restless, anger	3 / 3 / 2	Sama Prakriti (Sannipataja / Tridoshaja)	C3	Pass
T5	thin, dry, irregular, light, dislikes_cold, quick_anxious, restless, worry (all Vata options)	8 / 0 / 0	Vata Prakriti (Ekadoshaja)	C1	Pass
T6	large, oily, slow, deep, dislikes_damp, calm_steady, slow_steady, withdrawal (all Kapha options)	0 / 0 / 8	Kapha Prakriti (Ekadoshaja)	C1	Pass
T7	thin, dry, irregular, light, dislikes_cold, sharp_irritable, purposeful, anger	5 / 3 / 0	Vata Prakriti (Ekadoshaja)	C1 (top - mid = 2)	Pass
T8	thin, dry, irregular, light, dislikes_heat, sharp_irritable, purposeful, withdrawal	4 / 3 / 1	Vata-Pitta Prakriti (Dvandvaja)	C2 (top - mid = 1, top - low = 3)	Pass
T9	thin, dry, irregular, light, dislikes_heat, sharp_irritable, purposeful, anger	4 / 4 / 0	Vata-Pitta Prakriti (Dvandvaja)	C2 (tie at top)	Pass
T10	thin, dry, irregular, moderate, dislikes_heat, sharp_irritable, slow_steady, withdrawal	3 / 3 / 2	Sama Prakriti (Sannipataja / Tridoshaja)	C3 (top - low = 1)	Pass
T11	Enter 0, 4, text or blank at a question	N/A	Program rejects the input, prints "Invalid input, try again." and asks again	-	Pass

7.2 Sample Output and Test Analysis

Sample Output – Test Case T3

```
                    --------------- RESULT ---------------
Vata  score : 0
Pitta score : 4
Kapha score : 4
Prakriti    : Pitta-Kapha Prakriti (Dvandvaja)
Advice      : Prefer cooling, mildly sweet and bitter foods; avoid excess
              spicy, fried and sour food; avoid overheating and overwork. Also:
              Prefer light, warm, spiced food; take regular vigorous exercise;
              avoid heavy, oily, sweet food and daytime sleep.

----- EXPLANATION (rules fired) -----
R03: IF frame = large THEN Kapha +1
R06: IF skin = oily THEN Kapha +1
R08: IF appetite = strong THEN Pitta +1
R11: IF sleep = moderate THEN Pitta +1
R15: IF climate = dislikes_damp THEN Kapha +1
R17: IF temperament = sharp_irritable THEN Pitta +1
R21: IF activity = slow_steady THEN Kapha +1
R23: IF stress = anger THEN Pitta +1
C2: classification rule fired -> Pitta-Kapha Prakriti (Dvandvaja)
```

Testing conclusion: T1–T4 are representative cases of single-dosha, dual-dosha and balanced results. T5–T6 check the extreme scores (8 / 0 / 0). T7–T10 check the decision boundaries of C1, C2 and C3. T11 checks the input-validation loop in ask_user(). All 11 test cases passed.

CHAPTER 8: ADVANTAGES, LIMITATIONS AND FUTURE SCOPE

8.1 Advantages

- Simple, fast and easy to understand; no training data needed.
- Gives the same answer every time for the same input (consistent).
- Transparent: the explanation shows the reasoning.
- Rules can be edited or added without changing the inference engine.
- Helps students and beginners learn Prakriti classification.

8.2 Limitations

- Uses only eight attributes; a real Vaidya examines many more factors, including pulse (Nadi Pariksha), tongue and detailed history.
- All attributes have equal weight, which is a simplification.
- Answers are self-reported, so accuracy depends on the honesty and self-awareness of the user.
- The system classifies Prakriti only and cannot diagnose or treat illness.
- The text-based interface is basic.

8.3 Future Scope

- Add more attributes and weighted rules prepared with the help of an Ayurvedic practitioner.
- Add Vikriti (current imbalance) assessment along with Prakriti.
- Use fuzzy logic to handle in-between answers.
- Build a web or mobile application with a graphical interface and support for regional languages.
- Extend the classification to foods (by rasa and guna) and medicinal plants.
- Compare the rule-based results with a machine learning model trained on expert-labelled data.

8.4 Disclaimer

This project is made only for academic and educational purposes. It is not a medical device and its output must not be treated as medical advice. A qualified Ayurvedic practitioner or doctor should be consulted for any health concern.

CHAPTER 9: CONCLUSION

This project successfully demonstrated how the Ayurvedic concept of Prakriti classification can be modelled as a rule-based expert system. The classical knowledge about Vata, Pitta and Kapha was converted into 24 scoring rules and 3 classification rules, and a forward-chaining inference engine was used to classify a person into one of the seven Prakriti types. The explanation facility makes the reasoning visible and the system gave correct results in all test cases.

The work shows that the Indian Knowledge System is structured and logical enough to be combined with modern Artificial Intelligence methods. With more detailed rules prepared by domain experts and a better interface, such a system can become a useful educational and awareness tool.

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APPENDIX A: SOURCE CODE

File: `expert_system.py`

```
"""
Ayurvedic Classification as a Rule-Based Expert System Prakriti
(body constitution) classification using forward chaining.
Educational project - not a medical diagnosis tool.
"""

DOSHAS = ["Vata", "Pitta", "Kapha"]

# ---------------------- KNOWLEDGE BASE: attributes ----------------------
# attribute -> (question, [(value, description), ...])
# option order = Vata, Pitta, Kapha
ATTRIBUTES = {
    "frame": ("Body frame",
              [("thin", "Thin and light"),
               ("medium", "Medium and muscular"),
               ("large", "Large and heavy")]),
    "skin": ("Skin type",
             [("dry", "Dry and rough"),
              ("warm", "Warm, sensitive, prone to redness"),
              ("oily", "Oily, smooth and soft")]),
    "appetite": ("Appetite and digestion",
                 [("irregular", "Irregular appetite"),
                  ("strong", "Strong appetite, quick digestion"),
                  ("slow", "Low appetite, slow digestion")]),
    "sleep": ("Sleep pattern",
              [("light", "Light, easily disturbed"),
               ("moderate", "Moderate and sound"),
               ("deep", "Deep and long")]),
    "climate": ("Climate dislike",
                [("dislikes_cold", "Dislikes cold and wind"),
                 ("dislikes_heat", "Dislikes heat and sun"),
                 ("dislikes_damp", "Dislikes damp, cool weather")]),
    "temperament": ("Temperament",
                    [("quick_anxious", "Quick, enthusiastic, anxious"),
                     ("sharp_irritable", "Sharp, focused, irritable"),
                     ("calm_steady", "Calm, steady, forgiving")]),
    "activity": ("Activity level",
                 [("restless", "Restless, always moving"),
                  ("purposeful", "Purposeful and goal oriented"),
                  ("slow_steady", "Slow, steady, enduring")]),
    "stress": ("Reaction to stress",
               [("worry", "Worry and fear"),
                ("anger", "Anger and frustration"),
                ("withdrawal", "Withdrawal and silence")]),
}

ADVICE = {
    "Vata": "Prefer warm, cooked, moist foods; keep a regular routine; "
            "get enough rest; avoid cold, dry and very irregular habits.",
    "Pitta": "Prefer cooling, mildly sweet and bitter foods; avoid excess "
             "spicy, fried and sour food; avoid overheating and overwork.",
    "Kapha": "Prefer light, warm, spiced food; take regular vigorous "
             "exercise; avoid heavy, oily, sweet food and daytime sleep.", }


def build_rules():
  """Scoring rules R01..R24: IF attribute = value THEN dosha +1"""
rules, n = [], 1
for attr, (_, options) in ATTRIBUTES.items():
for i, (value, _) in enumerate(options):
            rules.append({"id": "R%02d" % n, "attr": attr,
"val": value, "dosha": DOSHAS[i]})
n += 1
```

```
    return rules

RULES = build_rules()

# ---------------- INFERENCE ENGINE (forward chaining) ----------------
def infer(facts):
    score = {d: 0 for d in DOSHAS}
    trace = []
    # Phase 1: fire every scoring rule whose condition matches the facts
    for r in RULES:
        if facts.get(r["attr"]) == r["val"]:
            score[r["dosha"]] += 1
            trace.append("%s: IF %s = %s THEN %s +1"
                         % (r["id"], r["attr"], r["val"], r["dosha"]))
    # Phase 2: classification rules on the accumulated scores
    ranked = sorted(DOSHAS, key=lambda d: -score[d])
    top, mid, low = (score[d] for d in ranked)
    if top - mid >= 2:
        rule, result = "C1", "%s Prakriti (Ekadoshaja)" % ranked[0]
        advice = ADVICE[ranked[0]]
    elif top - low <= 1:
        rule, result = "C3", "Sama Prakriti (Sannipataja / Tridoshaja)"
        advice = "Maintain balance of all three doshas with a varied diet."
    else:
        rule = "C2"
        result = "%s-%s Prakriti (Dvandvaja)" % (ranked[0], ranked[1])
        advice = ADVICE[ranked[0]] + " Also: " + ADVICE[ranked[1]]
    trace.append("%s: classification rule fired -> %s" % (rule, result))
    return {"score": score, "result": result,
            "advice": advice, "trace": trace}


def ask_user():
    facts = {}
    for attr, (title, options) in ATTRIBUTES.items():
        print("\n%s?" % title)
        for i, (_, desc) in enumerate(options, 1):
            print("  %d. %s" % (i, desc))
        while True:
            ch = input("Enter choice (1-3): ").strip()
            if ch in ("1", "2", "3"):
                facts[attr] = options[int(ch) - 1][0]
                break
            print("Invalid input, try again.")
    return facts


def show(out):
    print("\n----- RESULT -----")
    for d in DOSHAS:
        print("%-6s score: %d" % (d, out["score"][d]))
    print("Prakriti :", out["result"])
    print("Advice   :", out["advice"])
    print("\n----- EXPLANATION (rules fired) -----")
    for line in out["trace"]:
        print(line)


if __name__ == "__main__":
    print("AYURVEDIC PRAKRITI EXPERT SYSTEM")
    show(infer(ask_user()))
    print("\nNote: educational use only; consult a qualified Vaidya.")
```